

SHRI G.S. INSTITUTE OF TECHNOLOGY AND SCIENCE, INDORE - 3

Name : HARSH RAJ SHARMA

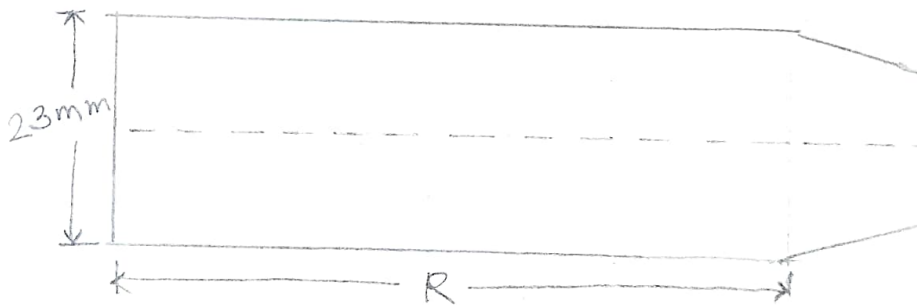
MACHINE-SHOP

Roll No. : AB-16030

TITLE : Machining the job as per drawing

- OBJECTIVES :
- : To familiarise with the lathe m/c.
 - : Parts of lathe m/c and their function.
 - : Various operation performed on lathe m/c.
 - : Various cutting tools and their material.
 - : Cutting speed, feed and depth of cut.

DRAWING:



Material M.S.

Raw material size

100 mm X 25 mm

All the dia menslon in m.m.

Machine/Equipment used	Particular	Specification
Hand Tool / Cutting Tool used	Bed Tool, Post single point, cutting tool	Made of high speed steel tool
Measuring / Working Tool used	Meter scale, vernier scale, cut side calliper	Steel Rule

Sequence of operations :

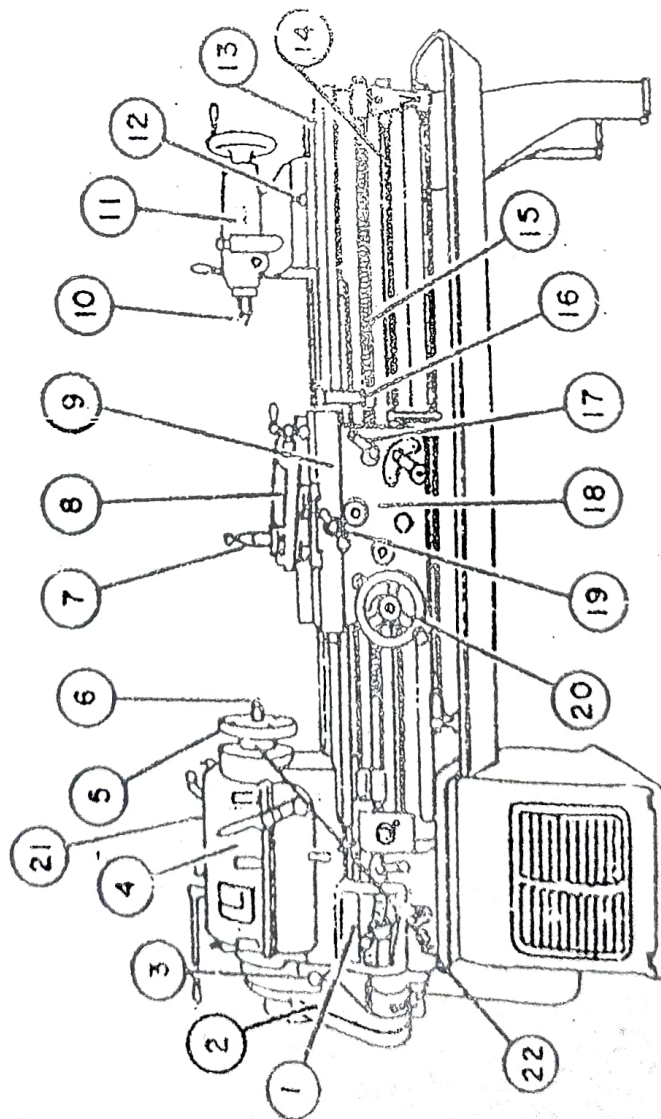
- 1.) First row of cylinder is fixed in chuck.
- 2.) Workpiece is set b/w chuck and lathe centre by adjusting steady to tail stock.
- 3.) Tool is fixed in tool post.
- 4.) Centering & surfacing of job is done.
- 5.) The job is reduced to required shape and dimensions.
- 6.) Taper Angle = $\frac{(\text{Big Diameter} - \text{Small diameter})}{\text{length of job}} \times \frac{200}{1}$

Precautions:

- 1.) Tools should be properly ground at correct height and properly set for turning.
- 2.) Revolving chuck should not be stopped by hand.
- 3.) Tools and instruments should not be played away lathe bed.
- 4.) Distance from revolving chuck should be maintained.

Title : THE CENTRE LATHE (WORK SHEET)

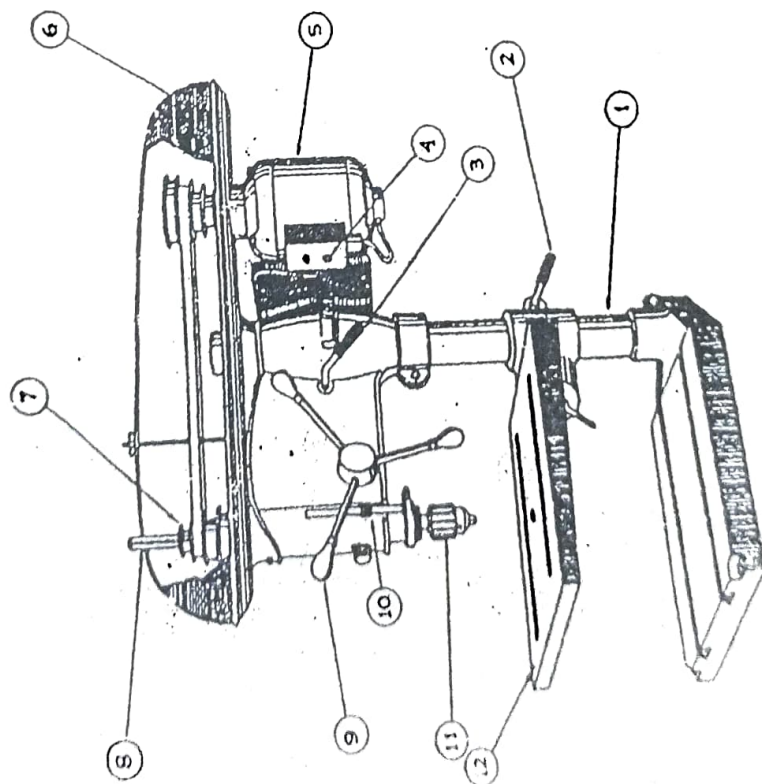
Direction : Write the number of each part opposite its proper name listed below.



Machine 4

- | | |
|-----|-----------------------|
| () | Tail Stock |
| () | Gear Guard |
| () | Head Stock |
| () | Drive Plate |
| () | Tool Plate |
| () | Tool Post |
| () | Live Centre |
| () | Dead Centre |
| () | Reverse Lever |
| () | Lead Screw |
| () | Back Gears |
| () | Compound Rest |
| () | Spindle Sleeve |
| () | Gear Box |
| () | Apron |
| () | Hand Wheel |
| () | Bed |
| () | Carriage |
| () | Set Over Screws |
| () | Half Nut Lever |
| () | Cross Feed Handle |
| () | Thread Dial Indicator |
| () | Feed Rod |

Title : THE BENCH DRILL (WORK SHEET)



Direction : Write the number of each part in the proper space

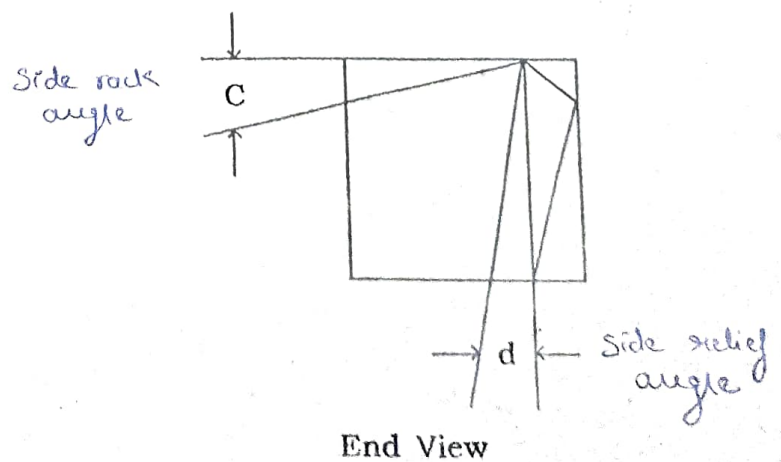
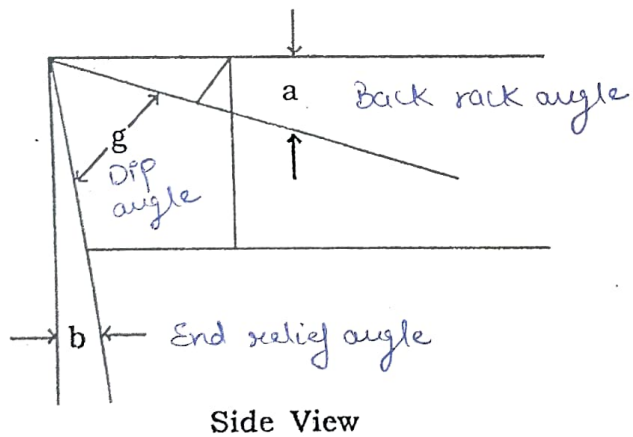
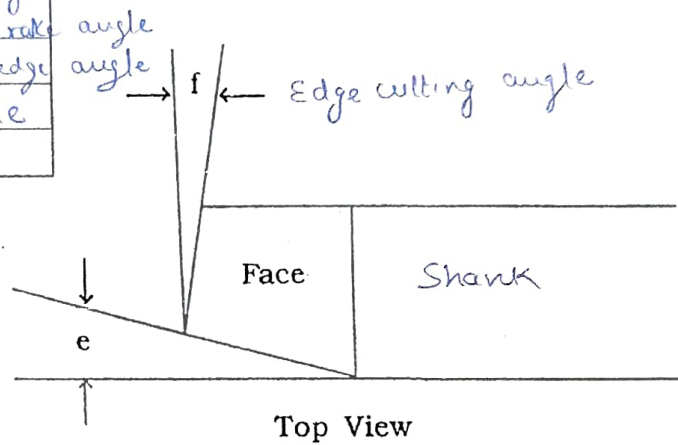
- () Adjustable Stop
- () Work Table
- () Guard
- () Coloum
- () Clamp for Head
- () Spindle
- () Drill Cluck
- () Start Stop Button
- () Speed Change Pulley
- () Hand Feed Wheel
- () Motor
- () Work Table Clamp

Direction : Write the number of part in the space opposite the statement of the function of the part.

- () To Support the Work
- () To Start and Stop the Motor
- () To Limit the Depth of the Hole
- () To Hold the Drill
- () To Change the Spindle Speed
- () To Support the Work Table and Head
- () To Turn the Drill
- () To Raise and Lower the Drill
- () To Clamps the Head in Place
- () To Encase the Pulleys and the Drill

Single Point Cutting Tool

Angle	Name
a	Back rake angle
b	End relief angle
c	Side rake angle
d	Side relief angle
e	side cutting rake angle
f	End cutting edge angle
g	Wedge angle



QUESTIONARIES :

Machine 3

- 01 What are different operations generally performed on lathe machine?
- 02 What are the different tool materials ?
- 03 What is the composition of H.S.S. tool ?
- 04 Write the function of following:
 - (a) Tool post
 - (b) Head stock
 - (c) Compound slide/ rest
 - (d) Lead screw
 - (e) Carriage
 - (f) Stepped cone pully
- 05 What are different types of lathe m/c ?
- 06 Give the specification of lathe m/c ?
- 07 What are the different method of Taper Turning ?
- 08 Explain the term cutting speed, feed and depth of cut ?
- 09 What are the different type of holding devices used on lathe m/c ?
- 10 What is function of "Back Gear" ?
- 11 complete the work: sheets as enclosed herewith ?
 - (a) The center lathe
 - (b) Drill m/c
 - (c) Single point cutting tool.

1.) The different operations generally performed on a lathe machine are:-

- (i.) cylindrical turning (ii.) Taper turning (iii.) Boring
(iv.) Facing (v.) Thread cutting (vi.) Drilling
(vii.) ~~Reaming~~ ^{Reaming} (viii.) Tapping (ix.) Knurling
(x.) Putting off

2.) The different tool material are:-

- (i.) Carbon steel (ii.) Medium alloy steel
(iii.) High speed steel (iv.) Mild steel (v.) Alurines
(vi.) Stellite (vii.) Diamond cutting tools

3.) The composition of HSS is

- (i.) Holypotenum HSS (ii.) (a) Tungsten:- 6%
(b) Chromium:- 4% (c.) Molybdenum:- 6%
(d) Vanadium:- 2%
(iii.) Cobalt.
(iv.) (a) Tungsten - 20% (b) Vanadium:- 4% (c) Cobalt:- 2%

4.) (i) Tool post - It is located at the top of the compound rest to hold the tool and to allow it to adjust to a consistent working point.

(ii) Head stock - The head stock is secured permanently on the innerways at the left hand of lathe bed & it provides mechanical means of rotating the work piece.

(iii) Compound slide/rest - It is mounted on the top of the cross slide and has a circular bed base graduated in degrees. It is used to cut angular cuts.

(iv) Dead screw - It is a long threaded shaft used as master screw and is brought into operations only when threads have to be cut.

(v) Carriage - It is a tool supporting & grinding device. It is mounted on the lathe bed and travel along its length. It consists of three main parts: - saddle, apron, cross slide.

(vi) Stopped cone pulling - It is a mechanism for drawing a lathe. It is connected with motor, a hollow spindle running through stock which receive a rotating motion.

5) Different types of lathe machine are:-

(i) Bench lathe

(ii) Turret lathe

(iii) Engine lathe

(iv) Custom lathe

6) Lathe machine is specified by defining its various features as follows:-

(i) Maximum length b/w centers. It represents the max length of the job & can revolve without the bed.

- (ii) Diameter of bed and the carriage. It is the largest diameter of job that will revolve without touching bed.
- (iii) No of spindle, speed & their range
- (iv) Thread & pitch
- (v) Pitch of lead screw
- (vi) Drive motion power
- (vii) Depth of bed

7.) Different methods of taper turning are:-

- (i) By a broad, loose form a tool
- (ii) By setting over a tail stock.
- (iii) By checking turning attachment
- (iv) By combining longitudinal and cross feed in a special lathe

8.) Cutting speed:- It is the distance travelled by cutting tool point on the cutting edge relative to the surface of cut in unit time.

• Feed:- It is the amount of tool advancement for revolution of job parallel to the surface of cut in unit time being machined.

• Depth of cut- It is the thickness of layer of the metal removed in each cut, is measured in direction to surface being machined.

9.) The different types of holding devices are:-

- (i) Chuck
- (ii) Angle plate
- (iii) Face plate
- (iv) Rest

10-) Back gear is used to obtain range of spindle speed. It is also used & necessary to have a slow speed of the spindle that can't be otherwise obtained.